

Lendo Capital - Quant Research: Week 2 Summary

Factor Research: Value Proxy & Momentum | Aug 2021 - Jul 2025 | 51-ticker universe

1 Overview

Week 2 extended the Week 1 data pipeline into factor research, building and evaluating two cross-sectional equity signals -- a contrarian value proxy and a standard price-momentum factor -- across a 51-ticker universe over 48 monthly holding periods (Aug 2021 - Jul 2025). The pipeline is implemented in `src/factor_utils.py` and the consolidated deliverable is `notebooks/week2/factor_value_momentum.ipynb` (all 15/15 quality checks pass).

2 Scope & Data

Universe: 51 tickers -- top 50 S&P 500 by market cap as of 2025-08-30, plus SPY

Date rang

3 Factor Signals

Value proxy (contrarian trailing-return construction -- not fundamental P/B)

score = $-(\text{trailing 12-month adj_close return})$. High score = stock fell most = treated as 'cheaper.' No fundamental data available, so this is a proxy only. 2,498 signal records across 49 dates; PLTR excluded at 2021-08-31 (insufficient history).

Momentum (standard 12-1, Jegadeesh-Titman definition)

score = $P(t-1) / P(t-12) - 1$. Most-recent month skipped to avoid short-term reversal contamination. 2,498 signal records across 49 dates; same PLTR exclusion at 2021-08-31.

4 Quintile Portfolio Construction

- * At each rebalance date, tickers sorted descending by signal score and split into 5 buckets by rank (`np.array_split`).
- * Q5 = highest score (most desirable per signal); Q1 = lowest. Sizes: 11/10/10/10/10 for 51 tickers.
- * Forward returns attached via `shift(-1)` per ticker -- no look-ahead. Last date (2025-08-31) excluded (NaN fwd return).
- * Output: 240 rows per factor (48 dates x 5 quintiles), equal-weighted within each bucket.

5 Performance Results (48 monthly periods, rf = 0%)

5a Value Proxy (Q5 = cheapest / most fallen, Q1 = most expensive / most risen)

Quintile	Ann. Return	Ann. Vol	Sharpe	Max Drawdown
Q1 (expensive)	27.72%	23.32%	1.19	-25.2%
Q2	15.78%	17.45%	0.90	-25.2%
Q3	15.66%	16.98%	0.92	-14.7%
Q4	13.43%	18.49%	0.73	-27.4%
Q5 (cheap)	22.98%	23.95%	0.96	-27.9%
Q5 - Q1 spread	-4.75%	25.43%	-0.19	-46.5%

5b Momentum (Q5 = prior winners, Q1 = prior losers)

Quintile	Ann. Return	Ann. Vol	Sharpe	Max Drawdown
Q1 (losers)	25.20%	24.69%	1.02	-29.5%
Q2	13.28%	17.05%	0.78	-21.3%
Q3	15.36%	17.15%	0.90	-25.4%
Q4	12.06%	18.18%	0.66	-24.0%
Q5 (winners)	29.05%	22.61%	1.28	-19.2%
Q5 - Q1 spread	+3.85%	25.23%	+0.15	-38.7%

6 Year-by-Year Q5 - Q1 Spread

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Year	Value Q5-Q1	Momentum Q5-Q1	Note
2021	+16.5%	-9.3%	Partial year (Aug-Dec)
2022	-5.2%	+7.9%	
2023	-4.6%	-1.5%	AI theme: NVDA +170%, PLTR +340%
2024	-43.4%	+36.4%	
2025	+1.9%	-5.8%	

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7 Key Findings

- * Value's Q5-Q1 spread was -4.75%/year (INVERTED) -- the opposite of a classical value premium.
- * Momentum showed the correct direction: Q5 outperformed Q1 by +3.85%/year, Sharpe +0.15.
- * Cross-factor spread correlation: -0.93 (Pearson, 48 monthly obs) -- near-algebraic inverse by construction.
- * 2024 was the defining year: value spread -43.4%, momentum spread +36.4%, driven by NVDA and PLTR.
- * Both spread Sharpes (-0.19, +0.15) are statistically indistinguishable from zero at 48 observations.
- * All quintiles delivered positive absolute returns in most years -- substantial long-side market beta present.

8 Discussion

Why value's spread is inverted

Survivorship bias is the leading explanation. The universe is today's top-50 by market cap, applied retroactively. Every stock is in the sample because it appreciated enough to remain top-50 by 2025. High-momentum names (NVDA, PLTR) scored 'expensive' (Q1) under the value proxy and kept rising -- they were pre-selected for being long-run winners. The construction also ties value Q5 (cheap) to past losers, approximately shorting the momentum signal in a period dominated by AI-theme stocks.

Momentum's positive but weak spread

Winners outperformed losers in the correct direction, but with Sharpe 0.15 and sign flips in most years. Stripping 2024 (the AI-theme outlier, +36.4%), the remaining four periods average close to flat. Consistent with a real but noisy signal that requires more data and varied market regimes to confirm.

The -0.93 cross-factor correlation is construction-specific

value_score = -(trailing return) and momentum_score = +(trailing return), so the quintile portfolios are near-mirror images each month. A fundamental value factor (P/B, earnings yield) would not share this algebraic near-identity with momentum; the -0.93 should not be generalised beyond this proxy.

Factor-neutral does not equal market-neutral

Long-short Q5-Q1 portfolios cancel out factor exposure but carry non-trivial market beta. True market neutrality requires computing and separately hedging net beta to SPY -- a step not taken here.

9 Limitations

- * Survivorship bias (most important): universe retroactively applied; excludes all former top-50 names that since declined.
- * Small sample: 48 monthly observations over ~4 years; neither spread Sharpe is statistically significant.
- * No out-of-sample or parameter validation: one signal parameterisation tested, entirely in-sample.
- * Value signal is a proxy: findings cannot be read as evidence about value investing broadly.
- * PLTR: IPO 2020-09-30; excluded at first rebalance; monthly vol 1.48x daily-scaled (volatility clustering).

10 Deliverables

src/factor_utils.py

All factor functions: signals, quintile builder, perf stats, spread series

factor_val

11 Potential Week 3 Directions

- * Covariance matrix estimation (sample, shrinkage, factor-model implied).
- * Factor risk attribution: how much of EW/VW portfolio return is explained by momentum vs. value?
- * Add a third factor (e.g. low-volatility) to break the near-algebraic value-momentum degeneracy.

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- * Market-neutral spread: compute and hedge net SPY beta of the Q5-Q1 portfolios.
- * Multi-factor composite: blend value and momentum into a single signal score.
- * Out-of-sample / walk-forward backtest to reduce in-sample overfit concerns.
- * Point-in-time universe (Compustat/Bloomberg) to eliminate survivorship bias.